MEASUREMENT OF RESISTIVITY & DETERMINATION OF BAND-GAP USING FOUR-PROBE METHOD

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Abstract

The resistivity of metallic & semiconducting samples was measured using the Four-Probe method, and the temperature dependence of resistivity was analyzed. The resistivities at room temperature were found to be $(25.21 \pm 0.57)\ \Omega cm$ for n-type Si, $(21.25 \pm 0.43)\ \Omega cm$ for n-type Ge, and $(4.9 \pm 0.1) \times 10^{-5}\ \Omega cm$ for aluminum. The variation of resistivity with temperature for n-type Ge showed thermally activated behavior, and from the linear plot of $\ln\rho$ vs $\frac{1}{T}$, the band gap was determined to be $(0.73 \pm 0.01)\ eV$. The results demonstrate the effectiveness of the four probe method in minimizing contact resistance & characterizing materials.

Objective :-

i) To measure resistivity of semiconductors & metal at room temperature

ii) To measure resistivity of semiconductor as a function of temperature & determination of energy band gap.

Theory :-

The Four-Probe method is a standard technique used for accurate measurement of resistivity in semiconductors and metals. This method is advantageous as it overcomes the issues of contact resistance, which often plagues two-probe measurements & allows for accurate results across a wide range of sample shapes & resistivity values (from $\mu\Omega$ to $M\Omega$).

* Four-Probe Method Principle :-

In the setup, four sharp, collinear probes are placed on the flat surface of the material being tested. An equal distance (S) is maintained b/w adjacent probes. A constant current (I) is passed through the outer electrodes, while the floating potential (V) is measured across the inner pair of probes.

The floating potential V_f at a distance r from an electrode carrying current I in a material of resistivity ρ_0 is given by,

$$\boxed{V_f = \frac{\rho_0 I}{2\pi r}}$$

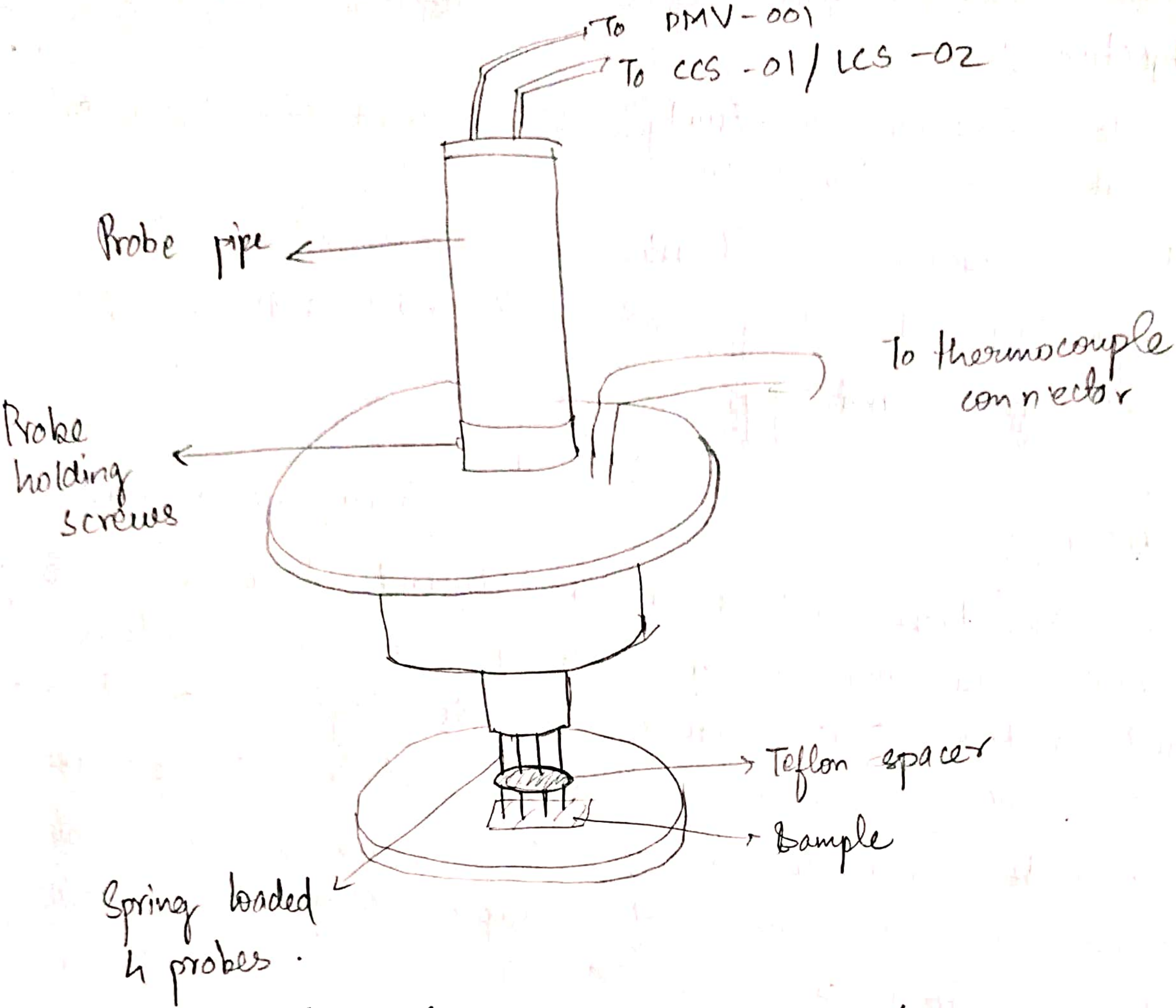

Fig: Four probe arrangement

When the point spacings are equal, i.e. $S_1 = S_2 = S_3 = S$, the resistivity of the material is given by:

$$\boxed{\rho_0 = \frac{V}{I} \times 2\pi S}$$

* **Correction for Thin Samples.**

When measurements are performed on thin slice of thickness W where $W \ll S$, the semi-infinite volume assumption fails & a correction factor

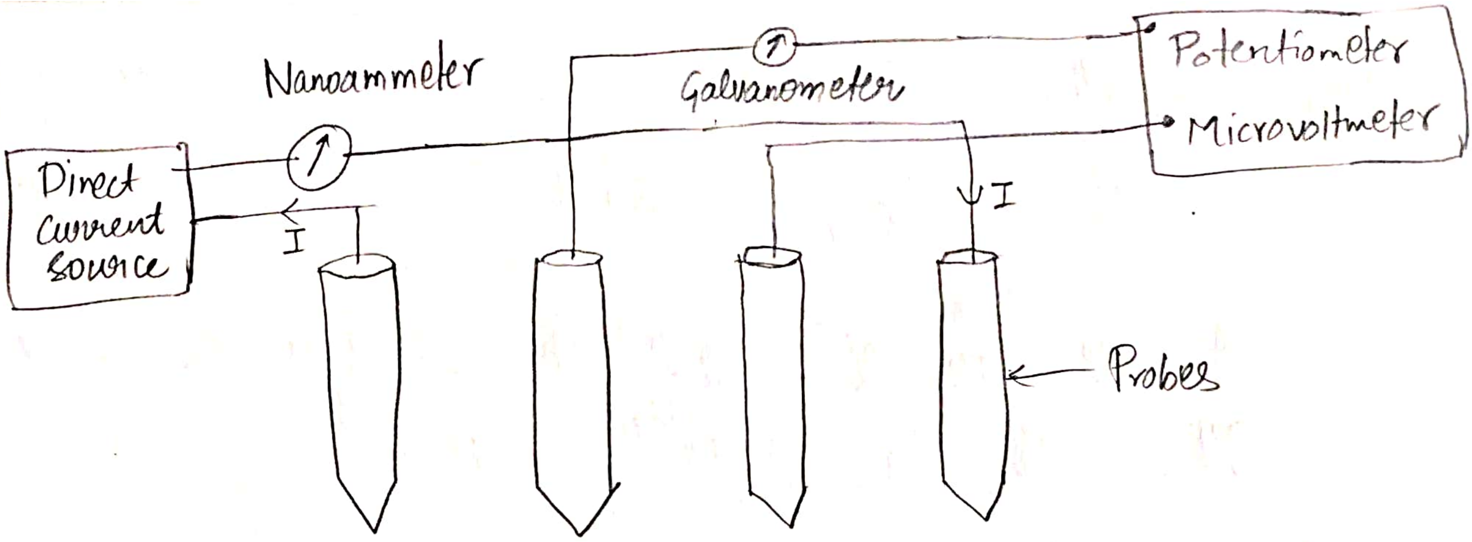

fig: Circuit used for resistivity measurement

$G(W/s)$ is applied,

$$\boxed{\rho = \frac{\rho_0}{G(W/S)}}$$

In cases, where sample rests on non-conducting bottom surface, the factor $G_7(W/s)$ is used. For very thin samples ($W/s < 0.25$), this correction factor simplifies leading to resistivity formula:

$$\boxed{\rho = \frac{V}{I}\frac{\pi W}{\ln(2)}}$$

* Determination of Energy Band gap :-

For a semiconductor, the resistivity (ρ) depends on the temperature (T) as per the following relation:

$$\boxed{\rho = A \exp\left(\frac{E_g}{2k_BT}\right)}$$

where,

E_g : band gap (in eV)

k_B : Boltzmann constant

T : Temperature

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Taking Logarithm, (base e)

$$\log \rho = \frac{E_g}{2k_B T} + \log A$$

$\therefore$ A linear fit of $\log \rho$ v/s $\frac{1}{T}$ gives the band gap energy E_g using the slope.

* Advantages of Four Probe Method :-

- Eliminates errors due to contact resistance
- Suitable for wide resistivity range.
- Minimizes heating & measurement errors.

* Assumptions behind Four probe method :-

- Probes are equally spaced and collinear.
- Contact area is small
- Sample is homogeneous and isotropic
- Surface is flat & free from leakage currents.

Apparatus Required :-

i) Four-probe arrangement (probes & sampe holder)
ii) PID Controlled Oven with thermocouple sensor
iii) Constant current source (for low resistivity samples)
iv) Low current source (for high resistivity samples)
v) Digital Microvoltmeter
vi) Semiconductor samples (n-type Si, Ge)
vii) Metal sample (Aluminium foil)
viii) Connecting wires

Observations :-

I = 4.8 mA

Thickness = 0.5mm   Res. = 18±1 Ω cm

Table :- n-Ge sample :-

S.no.	Temperature (T) (°C)	Voltage (V) (V)
1.	80°	0.20
2.	90°	0.170
3.	100°	0.152
4.	110°	0.145
5.	118°	0.055
6.	125°	0.047
11.	130	0.041
12.	135	0.039
13.	140	0.032
14.	145	0.029
15.	150	0.025
16.	155	0.022
17.	160	0.020
18.	165	0.018
19.	170	0.016
20.	175	0.014
21.	180	0.013
~~22.~~		

these readings were omitted.

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Observations :-

* Aluminium foil sample.   Thickness = 0.3 mm

S.no.	Temperature (T) (°C)	Voltage (V) (mV)
1.	27	0.060
2.	30	0.061
3.	35	0.063
4.	40	0.067
5.	45	0.060
6.	50	0.057
7.	55	0.069
8.	60	0.056
9.	65	0.057
10.	70	0.055
11.	75	0.064
12.	80	0.078

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Constant current ⇒ $I = 100$ mA.

W = thickness of the (Aluminum foil) sample = 0.3 mm

S = probe distance = 0.2 cm = ~~0.02 mm~~ 2 mm

$$\frac{W}{S} = 0.15$$

$$\cancel{G_7(W/S)} \quad G_7\left(\frac{W}{S}\right) = \frac{2S}{W}\log(2)$$

⇒ we use $\boxed{\rho = \frac{V}{I}\frac{\pi W}{\ln(2)}}$

Observations :-

Zero error in microvoltmeter = 0.7 mV

i) n-type Si

Res. = 22 ± 1 Ω cm

Thickness = 0.5 mm

S. no.	Current (I) (μA)	Voltage (V) (mV)	Corrected voltage ~~(V)~~ (mV)
1.	0	0.7	0
2.	10.2	1.9	1.2
3.	20.0	3.1	2.4
4.	30.0	4.4	3.7
5.	40.0	5.6	4.9
6.	50.4	6.9	6.2
7.	60.0	8.1	7.4
8.	70.5	9.4	8.7
9.	80.2	10.6	9.9
10.	90.3	11.9	11.2
11.	99.9	13.0	12.3
12.	110.0	14.3	13.6
13.	120.4	15.6	14.9
14.	130.0	16.8	16.1
15.	140.4	18.1	17.4
16.	150.0	19.3	18.6
17.	160.1	20.6	19.9
18.	170.0	21.8	21.1
19.	180.0	23.1	22.4

* Ge sample (n-type)

Res. = 18 ± 1 Ω cm
Thickness = 0.5 mm

S.no.	Current (I) (mA)	Voltage (V) (Volts)
1.	0.39	0.041
2.	0.90	0.088
3.	1.12	0.108
4.	1.42	0.136
5.	1.88	0.178
6.	2.20	0.207
7.	2.66	0.25~~6~~2
8.	3.33	0.314
9.	4.11	0.389
10.	4.39	0.414
11.	4.69	0.443
12.	5.04	0.475
13.	5.62	0.529
14.	6.02	0.568
15.	6.19	0.585

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Observations :-

* Al foil Thickness = 0.3 mm

S.no.	Current (I) (mA)	Voltage (V) (mV)
1.	4.0	0
2.	10.1	0.002
3.	21.1	0.006
4.	29.7	0.009
5.	40.5	0.014
6.	50.1	0.016
7.	60.0	0.020
8.	71.0	0.025
9.	80.4	0.028
10.	90.1	0.032
11.	100.4	0.036
12.	111.1	0.039
13.	122.3	0.044
14.	130.2	0.046
15.	140.8	0.050
16.	150.6	0.054
17.	160.0	0.056
18.	~~[illegible]~~ 169.7	0.060
19.	180.2	0.064
20.	190.1	0.067

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~~Dat~~ Calculations :-

(I) Calculations for Resistivity (ρ).

i) n-type Si sample:

Using the equation : $\rho = \frac{V}{I} \frac{\pi W}{\ln(2)}$

we plot V v/s I for each sample & obtain $\left(\frac{V}{I}\right)$ from the linear fit 's slope.

Here, we obtain slope = $a_1 = (124.49 \pm 0.14)\ \Omega$

we have, $\boxed{\rho = a_1 \cdot \frac{\pi W}{\ln(2)}}$

$W_{si\ sample} = 0.5\ mm$
(thickness)

$$\rho_{Si} = \frac{124.49 \times 3.14 \times 0.5 \times 10^{-3}}{\ln(2)}$$

$$\rho_{Si} = 0.2821\ \Omega m = 28.21\ \Omega cm$$

Similarly, we find resistivity for other samples,

ii) n-type Germanium sample :

From the linear fit we obtain, slope of V-I graph $= a_1 = (93.81 \pm 0.15)\ \Omega$

$$\rho_{Ge} = \frac{a_1 \pi W}{\ln 2}$$

$W_{Ge\ sample} = 0.5\ mm$
(thickness)

$$\rho_{Ge} = \frac{93.81 \times 3.14 \times 0.5 \times 10^{-3}}{\ln(2)} = 0.2125\ \Omega m = 21.25\ \Omega cm$$

iii) Aluminium foil sample :-

From the linear fit of V-I we obtain the slope,

$a_1 = (0.364 \pm 0.002) \times 10^{-3}\ \Omega$

$\rho = \frac{a_1 \pi W}{\ln(2)}$ $\qquad$ $W_{Al\ sample}$ (thickness) $= 0.3\ mm$

$\rho_{Al} = \frac{0.364 \times 3.14 \times 0.3 \times 10^{-3} \times 10^{-3}}{\ln(2)} = 0.4946 \times 10^{-6}\ \Omega m$

$= 0.4946 \times 10^{-4}\ \Omega cm = \underline{4.946 \times 10^{-5}\ \Omega cm}$.

~~= 4.946 Ω~~

(II) Calculations for Band Gap energy (E_g).

Using the equation: $\log \rho = \frac{E_g}{2k_B} \frac{1}{T}$ + constant

We perform a linear fit b/w $\log \rho$ (calculated using $\rho = \frac{V \pi W}{I \ln(2)}$) & $\frac{1}{T}$ (where T is in Kelvin) & obtain the slope of the least squares linear fit graph.

$a_1 =$ slope $= \frac{E_g}{2k_B}$ $\Rightarrow$ $E_g = 2k_B(\text{slope}) = 2k_B a_1$.

For n-type Ge sample, we obtain, $\qquad k_B = 8.617 \times 10^{-5}\ \frac{eV}{K}$

$a_1 = (4243.51 \pm 70.46)$ Kelvin

Band gap, $E_g = 2 \times 8.617 \times 10^{-5}\ \frac{eV}{K} \times 4243.51\ K$.

$\boxed{E_g \approx 0.731\ eV}$

The plot b/w resistivity & Temperature for Aluminium foil sample shows an overall increasing trend as expected, but has a lot of fluctuations to draw a meaningful conclusion from the data.

Error Analysis :-

(I) Linear fit : Least Squares Method :-

We fit the observed data points $\{x_i, y_i\}_{i=1}^{N}$ into a form

$$y = \overset{a}{\cancel{m}}x + b$$

a ~~m~~ = slope
b = intercept

~~The statistical error calculated in 1i~~

The optimal parameters m & c are obtained by minimizing the loss function.

$\hat{y}_i$ = true value

$$L(a, b) = \sum_{i=1}^{N} (\hat{y}_i - (ax_i + b))^2$$

$$\Rightarrow \frac{\partial L}{\partial a} = -2 \sum_{i=1}^{N} x_i (y_i - (ax_i + b))^2 = 0$$

N = no. of observations.

$$\frac{\partial L}{\partial b} = -2 \sum_{i=1}^{N} (y_i - (ax_i + b))^2 = 0$$

$$\Rightarrow \left\{ \begin{array}{l} a \sum_{i=1}^{N} x_i^2 + b \sum_{i=1}^{N} x_i = \sum_{i=1}^{N} x_i y_i \\ a \sum_{i=1}^{N} x_i + bN = \sum_{i=1}^{N} y_i \end{array} \right\}$$

& 2 variables.

Solving the linear system of 2 equations gives a & b.

Error in y, $\sigma_y = \sqrt{\frac{1}{(N-2)} \sum_{i=1}^{N} (y_i - (ax_i + b))^2}$

$\Delta = N \sum_{i=1}^{N} x_i^2 - \left(\sum_{i=1}^{N} x_i\right)^2$

Error in slope (a), $\sigma_a = \sigma_y \sqrt{\frac{N}{\Delta}}$

Error in intercept (b), $\sigma_b = \sigma_y \sqrt{\frac{\sum_{i=1}^{N} x_i^2}{\Delta}}$

(II) Propagation of Error :-

i) ~~Cal~~ Uncertainty in Resistivity (ρ).

We used the formula $\rho = \frac{V}{I}\frac{\pi W}{\ln(2)} = \frac{a_1 \pi W}{\ln 2}$

to find ρ.

~~As~~ Considering a $\pm 2\%$ error in thickness of the sample for all samples (as given in manual).

Uncertainity in ρ, $\Delta\rho = \sqrt{\left(\frac{\partial\rho}{\partial a_1}.\Delta a_1\right)^2 + \left(\frac{\partial\rho}{\partial W}.\Delta W\right)^2}$

$$\Delta\rho = \sqrt{\left(\frac{\pi W}{\ln(2)}.\Delta a_1\right)^2 + \left(\frac{a_1\pi}{\ln(2)}.\Delta W\right)^2} = \sqrt{\left(\frac{a_1\pi W}{\ln 2}.\frac{\Delta a_1}{a_1}\right)^2 + \left(\frac{a_1\pi W}{\ln(2)}.\frac{\Delta W}{W}\right)^2}$$

$$\boxed{\Delta\rho = \rho\sqrt{\left(\frac{\Delta a_1}{a_1}\right)^2 + \left(\frac{\Delta W}{W}\right)^2}}$$

$\frac{\Delta W}{W} \simeq 0.02$ (2%) relative error in thickness.

i) for n-type Si sample

$$\Delta\rho_{Si} = \rho_{Si}\sqrt{\left(\frac{0.14}{124.49}\right)^2 + (0.02)^2}$$

$$\approx 28.21\ (\Omega cm) \times (0.02)$$

$$\Delta\rho_{Si} = 0.57\ \Omega cm$$

ii) for n-type Ge sample

$$\Delta\rho_{Ge} = 21.25\ (\Omega cm)\sqrt{\left(\frac{0.15}{93.81}\right)^2 + (0.02)^2}$$

$$\Delta\rho_{Ge} = 0.43\ \Omega cm$$

iii) for Aluminum foil sample

$$\Delta\rho_{Al} = (4.946 \times 10^{-5}\ \Omega cm)\sqrt{\left(\frac{0.002}{0.364}\right)^2 + (0.02)^2}$$

$$\Delta\rho_{Al} = (0.1\ \cancel{\Omega cm} \times 10^{-5}\ \Omega cm)$$ ~~= 0.001 Ωcm~~

~~Results & Discussion~~ :-

iii) ii) Uncertainty in Band Gap (E_g).

We used the formula, $E_g = 2k_B \cdot a_1$ to find band gap energy where a_1 is the slope of $\log \rho$ v/s $\frac{1}{T}$ plot.

Uncertainty in E_g, $\Delta E_g = \sqrt{\left(\frac{\partial E_g}{\partial a_1} \cdot \Delta a_1\right)^2} = |2k_B \Delta a_1|$

$$\Delta E_g = \left|2k_B a_1 \frac{\Delta a_1}{a_1}\right| = E_g \left|\frac{\Delta a_1}{a_1}\right|$$

∴ Error in band gap,

$$\Delta E_g = 0.73 \text{ eV} \times \left(\frac{70.46}{4243.51}\right)$$

$$\boxed{\Delta E_g \approx 0.01 \text{ eV}}$$

Deviation from theoretical / Literature values.

i) Resistivity (ρ).

• For n-type Si,

$$\% \text{ dev} = \frac{|\rho_{measured} - \rho_{theoretical}|}{\rho_{theo.}} \times 100\% = \frac{|22 - 28.21|}{22} \times 100\%$$

$\approx 28\%$

($\rho_{theoretical}$ for Si (n-type) $= 22 \pm 1 \ \Omega cm$) (given on the sample)

• For n-type Ge,

$$\% \text{ dev.} = \frac{|21.25 - 18|}{18} \times 100\%$$

$\approx 18\%$

$\rho_{theoretical}$ for n-type Ge $= (18 \pm 1) \ \Omega cm$

• for Aluminum sample ~~we see that the orders of magnitude~~

$$\% \text{ dev} = \left(\frac{2.8 \times 10^{-6} - 0.49 \times 10^{-6}}{2.8 \times 10^{-6}}\right) \times 100\% = 1650\%$$

(off by an order of magnitude)

ii) Energy band gap (E_g)

$$\% \text{ deviation} = \frac{|0.68 - 0.73|}{0.68} \times 100\% \approx 7\%$$

Results & Discussion :-

The measured resistivities at room temperature using Four Probe method are :

- n-type Si :- $\rho_{Si} = (28.21 \pm 0.57)\ \Omega cm$
- n-type Ge :- $\rho_{Ge} = (21.25 \pm 0.43)\ \Omega cm$
- Aluminium :- $\rho_{Al} = (4.9 \pm 0.1) \times 10^{-5}\ \Omega cm$.

The Band gap of n-type Ge obtained from the $\ln\rho$ v/s $\frac{1}{T}$ plot is,

$$E_g = (0.73 \pm 0.01)\ eV$$

Compared with standard / literature values the resistivity of n-type Si & Ge have a deviation of 28% & 18% respectively. Although, for ~~al~~ Aluminium sample, the measured resistivity is off by an order of magnitude with a deviation of 1650%. The band gap calculated for n-type Ge was only 7% deviated from standard value. The $\ln\rho$ v/s $\frac{1}{T}$ plot for n-type Ge shows a linear trend confirming thermally activated conduction.

For aluminum, the ρ v/s T plot doesn't show a clear linear increase & exhibit large fluctuations. This is attributed to poor probe contact, sample inhomogeneity & measurement uncertainties which also caused the large deviation in resistivity calculation.

The higher deviation in n-type Si resistivity may be due to doping variation & contact issues, while relatively small deviation in Ge and its accurate band gap indicate better experimental accuracy for that part.

Conclusion

The four probe method was used to measure the resistivity of metallic & semiconducting samples & study their temperature variation. This method minimized contact resistance enabling accurate measurements. The experiment highlights the contrasting behaviour of metals & semiconductor, where Aluminium's resistivity increased with temperature, while n-type Ge exhibited thermally activated conduction, enabling determination of its band gap. Despite deviations arising from poor contact issues etc, the results confirm the effectiveness of Four Probe method in characterization of materials.

Sources of Errors :-

i) Improper contact of probe with sample

ii) Contamination of samples surface increases effective resistance

iii) Fluctuations in the current source. leads to improper measurements.

SAMPLE - n-type Silicon

Input Data

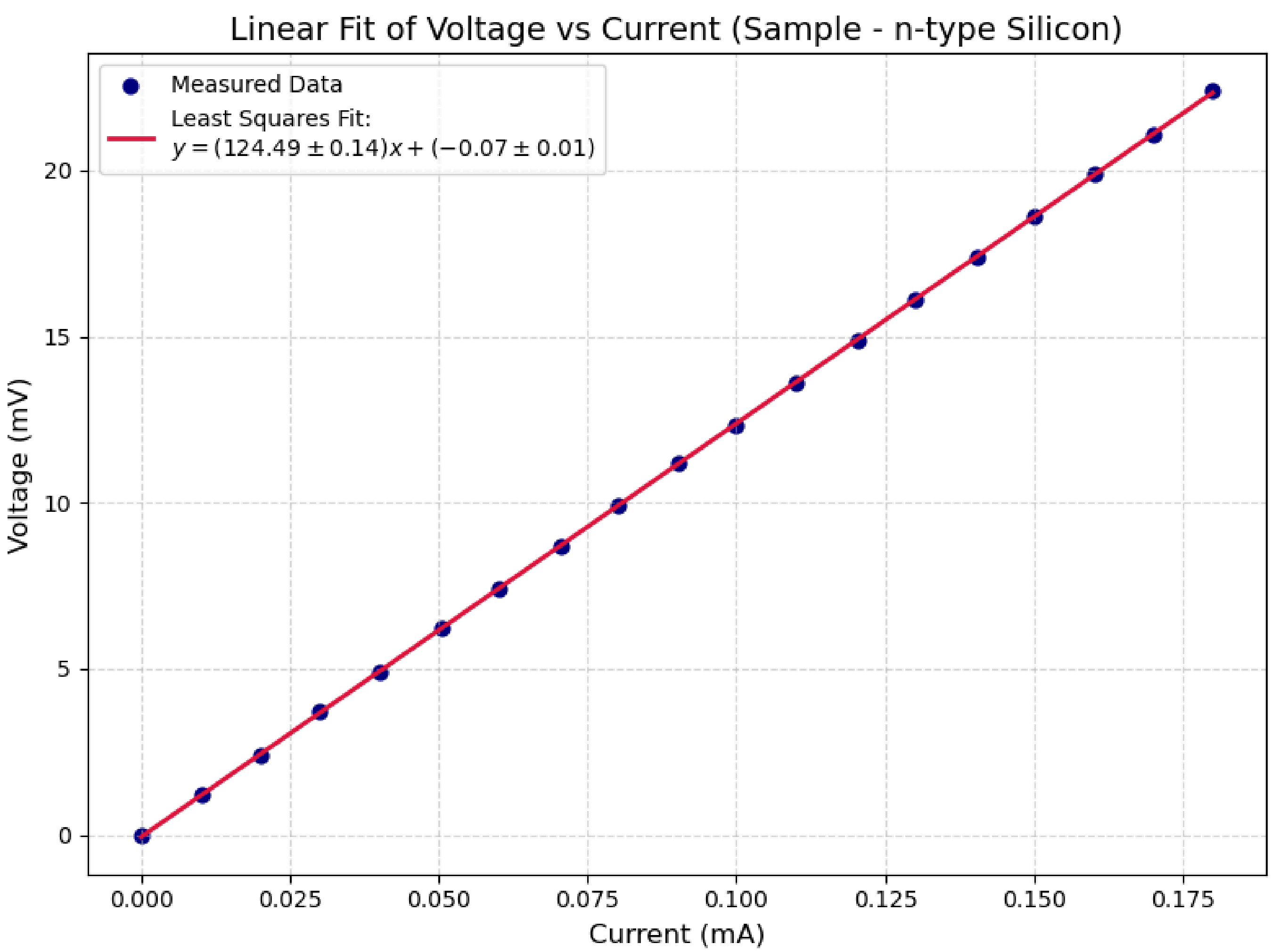
i	x	y	x^2	xy
1	0.0	0.0	0.0	0.0
2	0.01	1.2	0.0	0.012
3	0.02	2.4	0.0	0.048
4	0.03	3.7	0.001	0.111
5	0.04	4.9	0.002	0.196
6	0.05	6.2	0.003	0.312
7	0.06	7.4	0.004	0.444
8	0.07	8.7	0.005	0.613
9	0.08	9.9	0.006	0.794
10	0.09	11.2	0.008	1.011
11	0.1	12.3	0.01	1.229
12	0.11	13.6	0.012	1.496
13	0.12	14.9	0.014	1.794
14	0.13	16.1	0.017	2.093
15	0.14	17.4	0.02	2.443
16	0.15	18.6	0.022	2.79
17	0.16	19.9	0.026	3.186
18	0.17	21.1	0.029	3.587
19	0.18	22.4	0.032	4.032
Σ	1.712	211.9	0.211	26.192

Fit Results & Statistical Errors

Parameter	Value	Uncertainty
Slope (a_1)	124.49	±0.14
Intercept (a_0)	-0.07	±0.01
σ_y (Std. Error)	0.03	
Δ (delta)	1.08	

Measured vs Fitted Values

x	y (measured)	y (fit)
0.0	0.0	-0.067
0.01	1.2	1.203
0.02	2.4	2.423
0.03	3.7	3.668
0.04	4.9	4.913
0.05	6.2	6.207
0.06	7.4	7.402
0.07	8.7	8.709
0.08	9.9	9.917
0.09	11.2	11.174
0.1	12.3	12.369
0.11	13.6	13.627
0.12	14.9	14.921
0.13	16.1	16.116
0.14	17.4	17.411
0.15	18.6	18.606
0.16	19.9	19.863
0.17	21.1	21.096
0.18	22.4	22.341



SAMPLE - n-type Germanium

Input Data

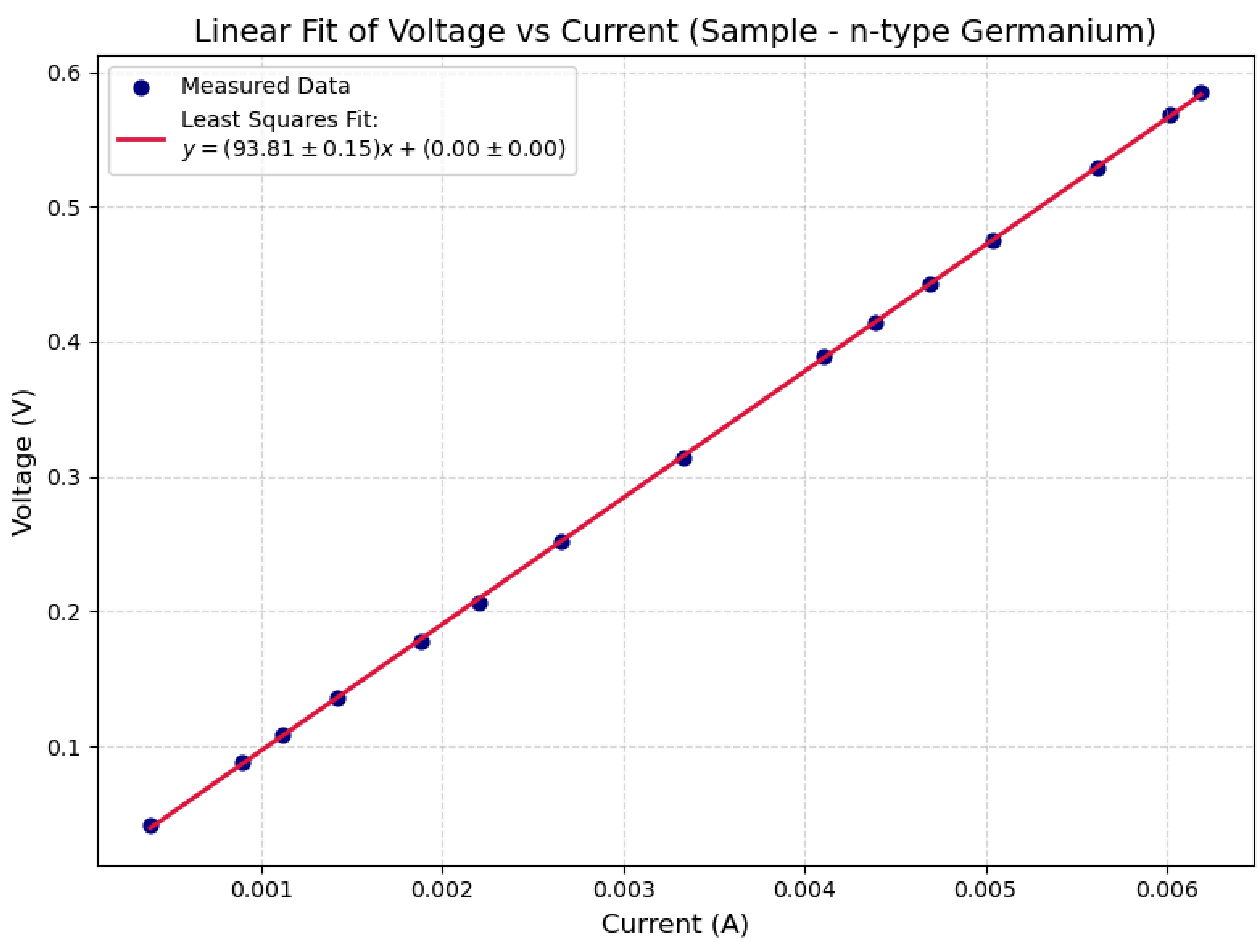
i	x	y	x^2	xy
1	0.00039	0.041	2e-07	2e-05
2	0.0009	0.088	8e-07	8e-05
3	0.00112	0.108	1.3e-06	0.00012
4	0.00142	0.136	2e-06	0.00019
5	0.00188	0.178	3.5e-06	0.00033
6	0.0022	0.207	4.8e-06	0.00046
7	0.00266	0.252	7.1e-06	0.00067
8	0.00333	0.314	1.11e-05	0.00105
9	0.00411	0.389	1.69e-05	0.0016
10	0.00439	0.414	1.93e-05	0.00182
11	0.00469	0.443	2.2e-05	0.00208
12	0.00504	0.475	2.54e-05	0.00239
13	0.00562	0.529	3.16e-05	0.00297
14	0.00602	0.568	3.62e-05	0.00342
15	0.00619	0.585	3.83e-05	0.00362
Σ	0.04996	4.727	0.00022	0.02082

Measured vs Fitted Values

x	y (measured)	y (fit)
0.00039	0.041	0.03928
0.0009	0.088	0.08712
0.00112	0.108	0.10776
0.00142	0.136	0.1359
0.00188	0.178	0.17905
0.0022	0.207	0.20907
0.00266	0.252	0.25222
0.00333	0.314	0.31507
0.00411	0.389	0.38824
0.00439	0.414	0.41451
0.00469	0.443	0.44265
0.00504	0.475	0.47548
0.00562	0.529	0.52989
0.00602	0.568	0.56741
0.00619	0.585	0.58336

Fit Results & Statistical Errors

Parameter	Value	Uncertainty
Slope (a_1)	93.81	±0.15
Intercept (a_0)	0.00	±0.00
σ_y (Std. Error)	0.00	
Δ (delta)	0.00	



SAMPLE - Aluminium foil

Input Data

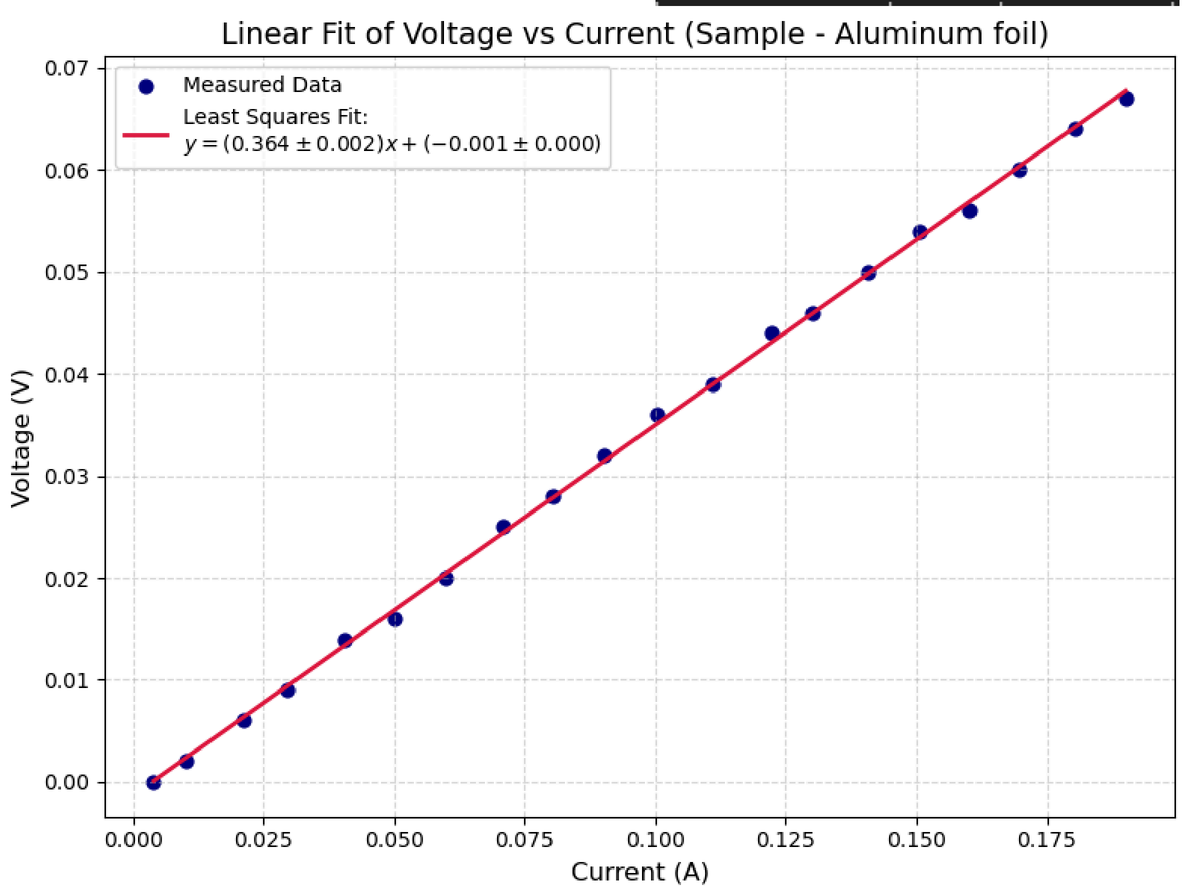
i	x	y	x²	xy
1	0.004	0.0	0.0	0.0
2	0.0101	0.002	0.0001	0.0
3	0.0211	0.006	0.0004	0.0001
4	0.0297	0.009	0.0009	0.0003
5	0.0405	0.014	0.0016	0.0006
6	0.0501	0.016	0.0025	0.0008
7	0.06	0.02	0.0036	0.0012
8	0.071	0.025	0.005	0.0018
9	0.0804	0.028	0.0065	0.0023
10	0.0901	0.032	0.0081	0.0029
11	0.1004	0.036	0.0101	0.0036
12	0.1111	0.039	0.0123	0.0043
13	0.1223	0.044	0.015	0.0054
14	0.1302	0.046	0.017	0.006
15	0.1408	0.05	0.0198	0.007
16	0.1506	0.054	0.0227	0.0081
17	0.16	0.056	0.0256	0.009
18	0.1697	0.06	0.0288	0.0102
19	0.1802	0.064	0.0325	0.0115
20	0.1901	0.067	0.0361	0.0127
Σ	1.9124	0.668	0.2487	0.0878

Measured vs Fitted Values

x	y (measured)	y (fit)
0.004	0.0	0.0001
0.0101	0.002	0.0023
0.0211	0.006	0.0063
0.0297	0.009	0.0094
0.0405	0.014	0.0134
0.0501	0.016	0.0169
0.06	0.02	0.0205
0.071	0.025	0.0245
0.0804	0.028	0.0279
0.0901	0.032	0.0314
0.1004	0.036	0.0351
0.1111	0.039	0.039
0.1223	0.044	0.0431
0.1302	0.046	0.046
0.1408	0.05	0.0498
0.1506	0.054	0.0534
0.16	0.056	0.0568
0.1697	0.06	0.0603
0.1802	0.064	0.0641
0.1901	0.067	0.0677

Fit Results & Statistical Errors

Parameter	Value	Uncertainty
Slope (a_1)	0.364	±0.002
Intercept (a_0)	-0.001	±0.000
σ_y (Std. Error)	0.001	
Δ (delta)	1.316	



Temperature Variation
SAMPLE - n-type Germanium

Input Data

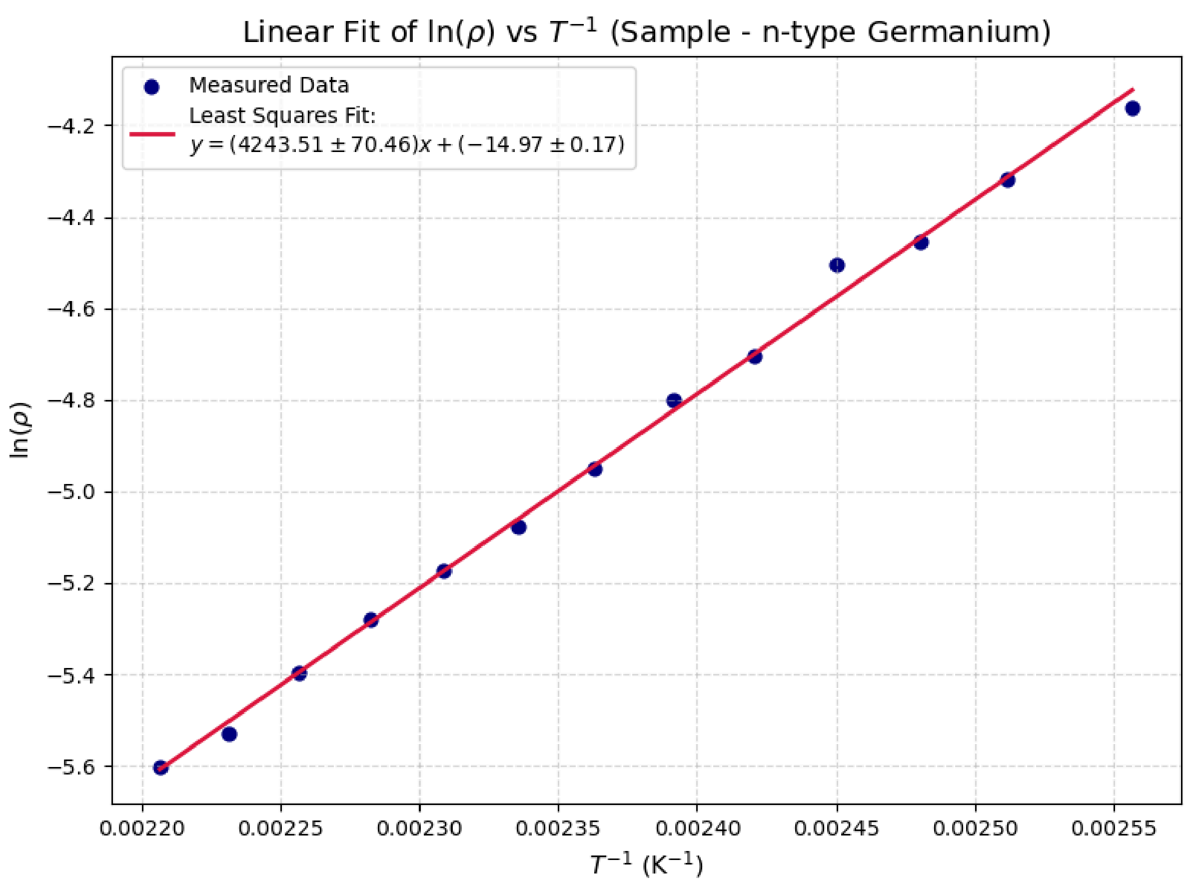
i	x	y	x^2	xy
1	0.00256	-4.35101	6.5e-06	-0.01112
2	0.00251	-4.5082	6.3e-06	-0.01132
3	0.00248	-4.64477	6.2e-06	-0.01152
4	0.00245	-4.69478	6e-06	-0.0115
5	0.00242	-4.89261	5.9e-06	-0.01184
6	0.00239	-4.99105	5.7e-06	-0.01194
7	0.00236	-5.13947	5.6e-06	-0.01215
8	0.00234	-5.2673	5.5e-06	-0.0123
9	0.00231	-5.36261	5.3e-06	-0.01238
10	0.00228	-5.46797	5.2e-06	-0.01248
11	0.00226	-5.58575	5.1e-06	-0.0126
12	0.00223	-5.71929	5e-06	-0.01276
13	0.00221	-5.79339	4.9e-06	-0.01278
Σ	0.0308	-66.4182	7e-05	-0.15671

Measured vs Fitted Values

x	y (measured)	y (fit)
0.00256	-4.35101	-4.31259
0.00251	-4.5082	-4.50332
0.00248	-4.64477	-4.63551
0.00245	-4.69478	-4.76445
0.00242	-4.89261	-4.89028
0.00239	-4.99105	-5.01309
0.00236	-5.13947	-5.13301
0.00234	-5.2673	-5.25012
0.00231	-5.36261	-5.36453
0.00228	-5.46797	-5.47633
0.00226	-5.58575	-5.5856
0.00223	-5.71929	-5.69244
0.00221	-5.79339	-5.79692

Fit Results & Statistical Errors

Parameter	Value	Uncertainty
Slope (a_1)	4243.51	±70.46
Intercept (a_0)	-15.16	±0.17
σ_y (Std. Error)	0.03	
Δ (delta)	0.00	



Temperature Variation
SAMPLE - Aluminium foil